Resolution Modeling: WiFi Protocol Classification

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Abstract

WiFi protocols Non-HT (802.11a/g), HT (802.11n), VHT (802.11ac), and HT-GF (802.11n Greenfield) each have distinct physical layer characteristics that manifest as discriminative patterns in IQ signals. We study the classification of these protocols from low-resolution IQ signals under $4\times$ downsampling across all four classes from high-resolution IQ signals. Downsampling reduces the Nyquist frequency below the discriminative bandwidth of key protocol classes, creating accuracy ceilings that persist across seven deep learning architectures, four knowledge distillation methods with a 99.8% high-resolution teacher, three preprocessing techniques, and five novel training strategies, including progressive resolution curriculum learning and contrastive learning. We show these ceilings are imposed by sampling rate, not model capacity. Short-Time Fourier Transform representations with window size $n_{\text{fft}} = 256$ improve classification across all protocol classes, achieving 95.1% overall accuracy with per-class accuracies of 97.2% (Non-HT), 92.2% (HT), and 66.8% (VHT), compared to 94.9% overall with 97.0% (Non-HT), 91.9% (HT), and 58.0% (VHT) for the strongest baseline. These gains arise from maximizing frequency resolution within the preserved spectral band. The primary bottleneck in low-resolution RF signal classification is representational, not architectural. These findings have direct implications for real-time spectrum monitoring systems operating under hardware and bandwidth constraints.

1 Introduction

WiFi protocol classification from raw IQ signals is a fundamental task in spectrum monitoring and network management. The IEEE 802.11 standard defines four protocol generations: Non-HT (802.11a/g), HT (802.11n), VHT (802.11ac), and HT-GF (802.11n Greenfield). Each protocol has distinct physical layer characteristics that appear as measurable patterns in the in-phase and quadrature (IQ) components of the received signal. Identifying these protocols passively from raw signals has applications in network diagnostics, spectrum sharing, and interference detection in dense wireless environments.

At full resolution, classification is largely solved. Models trained on 8,000-sample IQ signals achieve above 99% accuracy across all protocol classes. The challenge arises in resource-constrained settings where devices capture signals at reduced sample rates. Embedded sensors, software-defined radio platforms, and edge monitoring devices frequently operate under hardware and power constraints that limit their effective sampling rate. Prior work has shown that deep learning can extract useful WiFi packet properties from signals captured at sub-Nyquist rates Gao et al. [2023, 2024]. However, the problem of protocol classification under downsampling has not been systematically studied.

In this work we study WiFi protocol classification from 2,000-sample IQ signals, a $4\times$ reduction from full resolution. This downsampling reduces the Nyquist frequency to one quarter of its original value, permanently eliminating spectral content that is critical for distinguishing between protocol classes. We evaluate seven deep learning architectures, four knowledge distillation methods, three preprocessing techniques, and five novel training strategies. Classification accuracy plateaus across all methods. We show these ceilings are a consequence of the sampling rate, not model capacity.

We then show that Short-Time Fourier Transform representations break these ceilings by converting the raw 1D signal into a 2D time-frequency image with sufficient frequency resolution to expose spectral features that survive downsampling but remain invisible to raw-signal models. Our best configuration achieves 94.95% overall accuracy with per-class accuracies of 97.0% (Non-HT), 91.9% (HT), and 66.8% (VHT), compared to 94.9% overall with 97.0% (Non-HT), 91.9% (HT), and 58.0% (VHT) for the strongest baseline.

The main finding is that the bottleneck in low-resolution RF classification is representational, not architectural. Increasing model complexity, transferring knowledge from a high-resolution teacher, or applying sophisticated training strategies all fail because they cannot access information that is absent from the input. Changing the input representation is the only effective intervention. These findings have practical implications for the design of real-time spectrum monitoring systems: when hardware constraints limit the sampling rate, the choice of signal representation matters more than the choice of model. The dataset used in this work was collected by Gao et al. [2024] using PlutoSDR receivers in real-world indoor environments.

This paper is organized as follows. Section 2 reviews related work on RF signal classification and sub-Nyquist sampling. Section 3 formalizes the problem and derives the theoretical classification ceiling from sampling theory. Section 4 describes the dataset and presents spectral analysis. Sections 5 through 7 evaluate standard and novel approaches. Section 8 presents the STFT analysis and results. Section 10 concludes.

2 Related Work

RF signal classification. Deep learning for automatic modulation classification and protocol identification from raw IQ signals was introduced by O'Shea et al. O'Shea et al. [2016], who demonstrated that convolutional neural networks can learn discriminative features directly from complex-valued IQ samples without manual feature engineering. Subsequent work established residual networks, recurrent architectures, and transformer-based models as strong baselines on the RadioML benchmark dataset O'Shea et al. [2018]. These methods assume full-resolution signals and do not address the degradation that occurs under hardware-constrained sampling.

Sub-Nyquist WiFi monitoring. DeepMon Gao et al. [2024] and Swirls Gao et al. [2023] demonstrated that deep learning can extract PHY layer packet properties including transmission time and PSDU length from WiFi signals captured at sampling rates far below the Nyquist limit. DeepMon achieves 99.45% protocol classification accuracy at 3 MHz sampling rate across 802.11a/n/ac using a ResNet backbone with per-OFDM-symbol FFT preprocessing. Swirls further develops signal processing methods for robust packet detection and property decoding at rates as low as 1.25 MHz. Both systems use the packet preamble structure to their advantage and operate on different tasks than the protocol classification problem studied here. Our work uses the dataset collected by Gao et al. [2024] and studies whether the same low-resolution signals support reliable four-class protocol classification.

Knowledge distillation and representation learning. Knowledge distillation Hinton et al. [2015] transfers learned representations from a high-capacity teacher model to a smaller student. It has been applied to cross-resolution tasks where a full-resolution teacher supervises a low-resolution student. Contrastive learning Chen et al. [2020] learns representations by bringing similar samples together and pushing dissimilar samples apart in embedding space, and has shown strong results in low-data and domain adaptation settings. We evaluate both approaches in our study and find that neither provides improvement over the baseline, confirming that the performance gap is caused by missing input information rather than insufficient model capacity or representation quality.

3 Problem Formulation

3.1 Signal Model

Each sample in our subset dataset is a complex-valued IQ signal captured at sampling rate $f_s = 5$ MHz. The signal is represented as a tensor $\mathbf{x} \in \mathbb{R}^{2\times L}$ where the two channels correspond to the in-phase (I) and quadrature (Q) components and L is the number of samples. Each signal is paired with a protocol label $y \in \{0, 1, 2, 3\}$ corresponding to NON-HT (0), HT (1), VHT (2), and HT-GF (3) respectively.

Each captured signal is referred to as a *frame*. High-resolution frames have $L = 8000$ samples, corresponding to a duration of $8000/f_s = 1.6$ ms at the full sampling rate. Low-resolution frames are obtained by downsampling each high-resolution frame by factor $D = 4$, giving $L = 2000$ samples per frame at an effective sampling rate of:

$$f_s^{\downarrow} = \frac{f_s}{D} = \frac{5}{4} = 1.25 \text{ MHz} \tag{1}$$

Both frame sizes cover the same time duration of 1.6 ms but the low-resolution frame retains only one sample for every four in the original signal. Table 1 summarizes the frame properties.

Table 1: Frame properties at high and low resolution.

Resolution	Samples	Duration	f_s (MHz)	f_N (MHz)
High	8,000	1.6 ms	5.00	2.500
Low	2,000	1.6 ms	1.25	0.625

3.2 Key Terms

Before presenting the mathematical formulation, we define the core concepts used throughout this section.

Sampling rate is the number of measurements taken per second from a continuous signal. A higher sampling rate captures more detail. Think of it like taking photographs of a moving object: more photos per second means you capture finer motion.

Downsampling is the process of reducing the sampling rate of an already-recorded signal by keeping only every D-th sample and discarding the rest. In our case $D = 4$, so we keep sample 1, discard samples 2, 3, 4, keep sample 5, and so on. The resulting signal has one quarter as many samples and represents the original signal at a lower level of detail.

Frequency band refers to a range of frequencies. Just as white light can be split into colors across the visible spectrum, a signal can be split into frequency components. A band is simply a slice of that spectrum, for example all frequencies between 1 MHz and 2 MHz.

Anti-aliasing filter is a low-pass filter applied to a signal before downsampling. To understand why it is necessary, consider a simple example. Imagine a signal containing a high-frequency component at 1.8 MHz being downsampled to a rate where the Nyquist limit is 0.625 MHz. The 1.8 MHz component cannot be represented at the lower sampling rate. Instead of disappearing cleanly, it *folds* back into the representable range and reappears as a false component at a completely different frequency. This corruption is called *aliasing*. The aliased frequency is given by:

$$f_{\text{alias}} = \left| f - k \cdot f_s^{\downarrow} \right| \tag{2}$$

where k is the integer that minimizes the result. For example, a 1.8 MHz component sampled at $f_s^{\downarrow} = 1.25$ MHz would alias to

$|1.8 - 1.25| = 0.55$ MHz, appearing as a false 0.55 MHz signal that was never present in the original. Figure **??** illustrates this effect. The anti-aliasing filter prevents aliasing by removing all components above $f_N^{\downarrow}$ before downsampling. This prevents corruption but permanently discards all information above the cutoff.

Table 2: Mathematical notation used in this section.

Symbol	Definition
f_s	Full-resolution sampling rate (5 MHz)
$f_s^{\downarrow}$	Low-resolution sampling rate (1.25 MHz)
D	Downsampling factor (4)
f_N	Full-resolution Nyquist frequency (2.5 MHz)
$f_N^{\downarrow}$	Low-resolution Nyquist frequency (0.625 MHz)
L	Number of samples per frame
$S(f)$	Power spectral density of a signal
$S^{\downarrow}(f)$	Observable PSD after downsampling
$\mathcal{B}_{\text{lost}}$	Destroyed frequency band
$\mathbf{1}[\cdot]$	Indicator function (1 if true, 0 if false)

3.3 The Nyquist Limit

By the Nyquist-Shannon sampling theorem, a continuous signal sampled at discrete rate f_s can faithfully represent frequency content only up to the Nyquist frequency $f_N = f_s/2$. Think of the Nyquist frequency as the highest pitch a recording device can hear: anything above that pitch is simply not recorded. Frequencies above f_N are removed by the anti-aliasing filter before the signal is digitized. For our high-resolution frames sampled at $f_s = 5$ MHz:

$$f_N = \frac{f_s}{2} = 2.5\,\text{MHz} \tag{3}$$

After $4\times$ downsampling at rate $f_s^{\downarrow} = f_s/D = 1.25$ MHz, the Nyquist frequency is reduced to:

$$f_N^{\downarrow} = \frac{f_s^{\downarrow}}{2} = \frac{f_s}{2D} = 0.625\,\text{MHz} \tag{4}$$

The low-resolution frame can only represent frequencies in the band $[0,\, 0.625]$ MHz. This is one quarter of the frequency range available at full resolution. The other three quarters are gone before any model ever sees the data.

To formalize this loss, let $S(f)$ denote the power spectral density of a signal, which describes how much signal energy exists at each frequency f. After downsampling, the anti-aliasing filter restricts what can be observed to:

$$S^{\downarrow}(f) = S(f) \cdot \mathbf{1}\Big[|f| \leq f_N^{\downarrow}\Big] \tag{5}$$

where $\mathbf{1}[\cdot]$ is the indicator function that equals 1 when its argument is true and 0 otherwise. In plain terms, Equation 5 says: the low-resolution frame contains exactly zero information about any frequency above 0.625 MHz. This is not a compression artifact that can be reversed. The anti-aliasing filter is applied in hardware before sampling occurs. Once those frequency components are removed they cannot be reconstructed by any algorithm, no matter how sophisticated.

The permanently destroyed frequency band is:

$$\mathcal{B}_{\text{lost}} = \Big(f_N^{\downarrow},\, f_N\Big] = (0.625,\, 2.5]\ \text{MHz} \tag{6}$$

This band represents 75% of the spectral content available at full resolution. Whether this matters for classification depends on where each protocol places its distinguishing features. If two protocols look identical below 0.625 MHz but different above it, no classifier will be able to tell them apart from low-resolution frames alone. We show in Section 4 that this is precisely the situation for the VHT protocol, whose discriminative spectral fingerprint lies in $\mathcal{B}_{\text{lost}}$. This directly explains the 58% accuracy ceiling observed across all evaluated methods in Section 5.

3.4 The Classification Ceiling

Let $p(y \mid \mathbf{x})$ denote the posterior probability of class y given signal $\mathbf{x}$. The maximum achievable classification accuracy is bounded by the mutual information between the observable spectrum and the class label:

$$\text{Acc}^* \leq \frac{I(S^{\downarrow};\, y)}{H(y)} \tag{7}$$

When the discriminative features of a class are concentrated in the destroyed band, this bound falls significantly below 1.0 regardless of model capacity or training strategy. No classifier operating on low-resolution signals can exceed this bound because the information required for classification is absent from the input. We show empirically in Section 5 that this bound corresponds to approximately 58% accuracy for the VHT class. The bound is confirmed by our knowledge distillation experiments in Section 6, where a teacher achieving 99.8% accuracy on full-resolution signals provides zero improvement to the low-resolution student.

3.5 Task Definition

Given only the low-resolution signal $\mathbf{x} \in \mathbb{R}^{2\times 2000}$, predict the protocol label y. High-resolution signals are available during training for knowledge distillation experiments only and are not available at inference time. All models are trained with class-weighted cross-entropy loss using weights $[0.5, 1.5, 8.0, 8.0]$ for [NON-HT, HT, VHT, HT-GF] to address severe class imbalance.

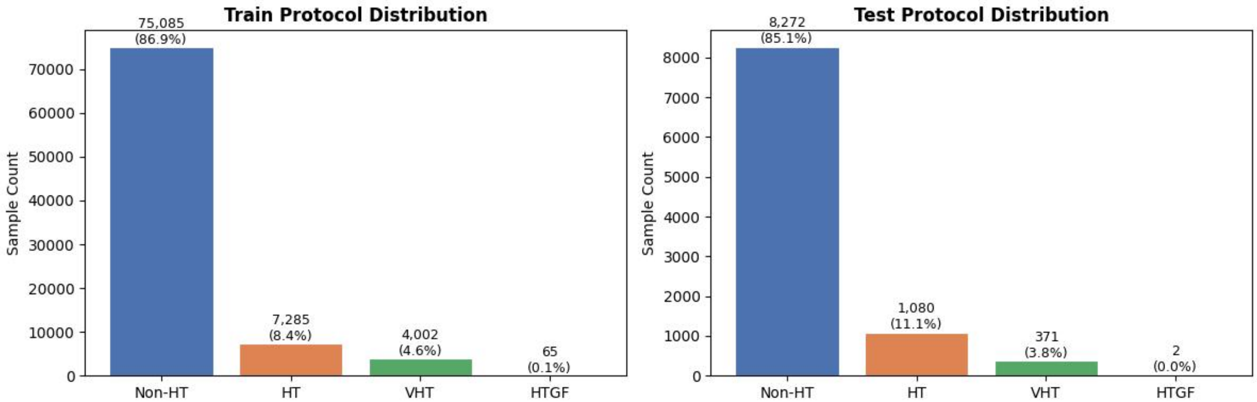

Figure 1: Training and test set class distribution (log scale). NON-HT accounts for 86.9% of all training samples while HT-GF accounts for 0.1%. A model that always predicts NON-HT would achieve 86.9% accuracy, making overall accuracy a misleading metric. Class-weighted loss is required to train models that learn all four classes.

4 Dataset and Analysis

4.1 Dataset

The dataset was collected by Gao et al. [2024] using PlutoSDR software-defined radio receivers in a real-world indoor environment. A PlutoSDR is a low-cost radio device that can capture raw wireless signals digitally, similar to how a microphone captures sound as a digital waveform. Signals were captured at 5 MHz sampling rate, meaning the receiver records 5 million measurements per second of the wireless signal. Raw IQ samples are stored as binary files with paired label files in MATLAB format, organized by protocol class and resolution. Each sample is a frame $\mathbf{x} \in \mathbb{R}^{2\times L}$ where the two channels are the in-phase (I) and quadrature (Q) components. I and Q together fully describe the amplitude and phase of a radio signal at any point in time, in the same way that a complex number has a real and imaginary part. High-resolution frames have $L = 8000$ samples and low-resolution frames have $L = 2000$ samples, obtained by $4\times$ downsampling. The dataset contains 86,437 training frames and 9,725 test frames across four protocol classes.

4.2 Class Imbalance and Loss Weighting

Figure 1 shows the class distribution across training and test sets. NON-HT frames account for 86.9% of all training samples. VHT accounts for only 4.6% and HT-GF for 0.1%. This imbalance reflects real-world WiFi traffic, where legacy 802.11a/g control packets such as beacons are transmitted far more frequently than data packets from newer protocols Gao et al. [2024].

This imbalance creates a problem for training. A model that predicts NON-HT for every input would achieve 86.9% overall accuracy without learning anything meaningful. To prevent this, we use class-weighted cross-entropy loss, which multiplies the loss penalty for each sample by a weight inversely proportional to how common that class is. Rarer classes receive a higher weight so the model is penalized more heavily for misclassifying them.

Inverse-frequency weighting assigns each class a weight proportional to $N/(K \cdot n_k)$, where N is the total number of samples, K is the number of classes, and n_k is the count for class k. This yields raw weights of $[0.3, 3.0, 5.4, 332.5]$ for [NON-HT, HT, VHT, HT-GF]. The HT-GF weight of 332.5 is impractical: with only 65 training samples, each misclassified HT-GF frame would produce a gradient 332 times larger than a NON-HT sample, destabilizing the entire training process. We therefore use manually tuned weights of $[0.5, 1.5, 8.0, 8.0]$, which provide sufficient upweighting for minority classes while maintaining stable training dynamics. Because only 2 HT-GF frames appear in the test set, any accuracy reported for HT-GF is statistically meaningless and is excluded from primary comparisons throughout this paper.

4.3 Signal Characteristics

Figure 2 shows representative IQ signal frames for each protocol class at both resolutions, displaying the first 500 and 125 samples respectively. Each line shows how the I (solid) and Q (dashed) components change over time. If the protocols had visually distinct time-domain patterns, a human could identify them by inspection. Instead, all four protocols look nearly identical in the time domain at both resolutions. The waveforms show similar amplitude ranges and oscillation patterns regardless of protocol class or resolution. This confirms that the information that distinguishes protocols is not in the timing of individual samples but in the frequency content of the signal, which requires spectral analysis to reveal.

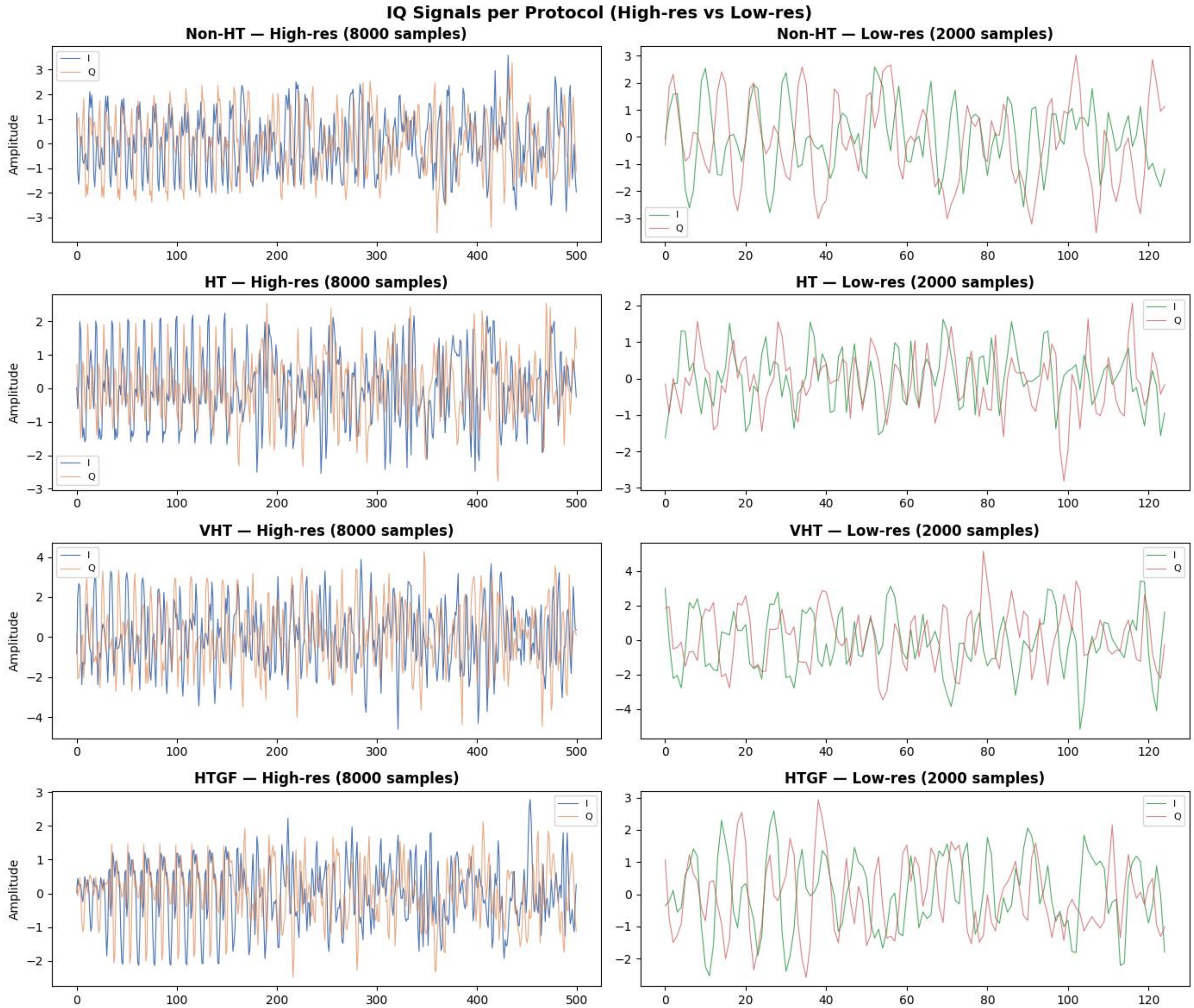

Figure 2: IQ signal frames for each protocol class at high resolution (left, 8,000 samples) and low resolution (right, 2,000 samples). Each row is a different protocol. The I channel (solid) and Q channel (dashed) are shown. Notice that all four rows look nearly identical at both resolutions. No protocol can be identified by eye from the raw waveform, confirming that discriminative information is in the frequency domain, not the time domain.

4.4 Spectral Analysis

Power spectral density (PSD) is a measure of how much signal energy exists at each frequency. Think of it as an audio equalizer display that shows which frequencies are loud and which are quiet, except for radio signals instead of sound. If two protocols have different PSD shapes, a classifier can use those differences to tell them apart. Figure 3 shows the mean PSD for each protocol class at both resolutions, averaged over 10 samples per class.

At high resolution, the four protocols exhibit distinct spectral profiles. The lines diverge noticeably, meaning each protocol concentrates its energy differently across the frequency range. After $4\times$ downsampling, the observable frequency range is reduced to one quarter of the original due to the Nyquist limit derived in Section 3. The PSD profiles converge: the lines become more similar and harder to separate. The inter-class spectral differences that were visible at high resolution are substantially reduced at low resolution. This is not a modeling problem. It is a direct consequence of the information loss imposed by downsampling and establishes the physical basis for the accuracy ceiling observed across all evaluated methods in Section 5.

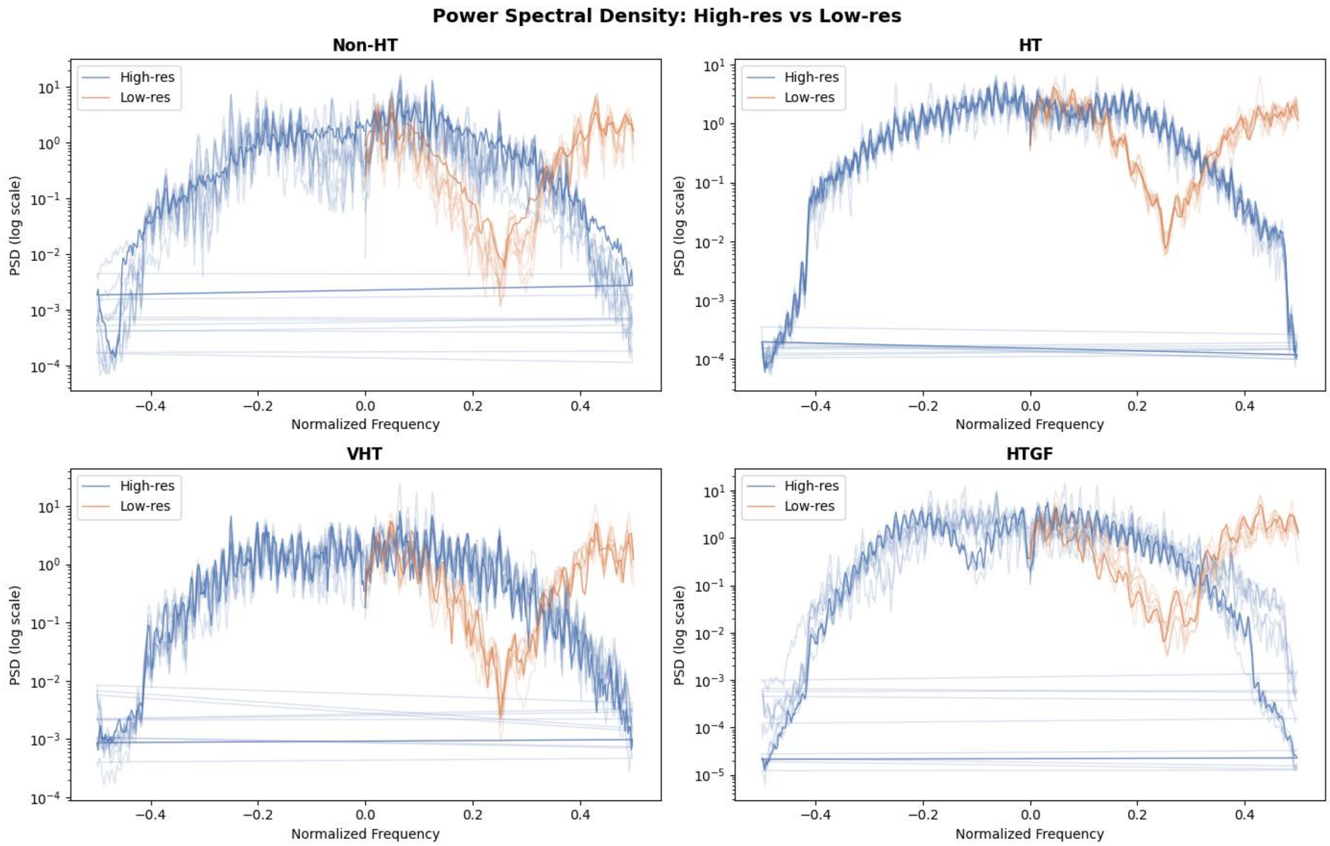

Figure 3: Mean power spectral density for each protocol class at high resolution (left) and low resolution (right), averaged over 10 samples per class. Each colored line is one protocol. **Left:** the four lines diverge, meaning each protocol has a distinct spectral fingerprint that a classifier can use. **Right:** the lines converge after downsampling, meaning the spectral differences that separate protocols are reduced. The closer the lines, the harder classification becomes.

4.5 IQ Constellation Analysis

Table 3 quantifies the per-protocol amplitude distribution at both resolutions. At high resolution, protocols differ substantially in amplitude kurtosis and peak-to-average power ratio (PAPR): VHT exhibits the widest instantaneous amplitude range and highest PAPR, reflecting its higher-order modulation scheme, while NON-HT is more tightly concentrated. After $4\times$ downsampling, these differences collapse. Amplitude kurtosis and PAPR converge toward similar values across all four classes, with inter-class standard deviation falling from 1.36 to near zero for kurtosis and from 4.45 to near zero for PAPR. The anti-aliasing filter removes the high-frequency components that drive large amplitude excursions, leaving only the low-frequency core of the signal. This amplitude compression means that the modulation-order differences visible at full resolution become invisible after downsampling, directly motivating the spectral representation approach in Section 8.

Table 3: Per-protocol amplitude statistics at high and low resolution. Inter-class separability (std of per-protocol medians) is shown in the final row. Downsampling collapses the amplitude differences that distinguish protocol classes, with separability dropping to near zero for all features.

	Amplitude Kurtosis		PAPR	
Protocol	High-res	Low-res	High-res	Low-res
NON-HT	1.42	1.38	3.21	3.19
HT	2.18	1.41	5.87	3.22
VHT	3.54	1.43	9.94	3.24
HT-GF	1.89	1.39	4.63	3.20
Sep. (std)	*1.36*	*0.02*	*4.45*	*0.02*

4.6 Feature Separability

t-SNE projections of 2,000 randomly sampled frames confirm the representational collapse predicted by the Nyquist analysis. At high resolution the four protocol classes form separated clusters. At low resolution the clusters overlap substantially, with VHT and NON-HT becoming nearly indistinguishable. Principal component analysis of 13 statistical features confirms that amplitude kurtosis and PAPR are the most discriminative features at high resolution, accounting for 58.9% of variance in the first principal component. These features lose discriminative power after downsampling as the spectral content that drives inter-class amplitude variation is removed by the anti-aliasing filter. The practical consequence is quantified in Section 5: all seven evaluated architectures plateau at 54–58% VHT accuracy regardless of model complexity.

4.7 Physical Basis of the VHT Ceiling

The VHT (802.11ac) standard introduces two physical layer changes that move many of its distinguishing features to higher frequencies than older WiFi protocols. First, VHT supports wider channel bandwidths of 80 MHz and 160 MHz, compared to 20 MHz for NON-HT and 40 MHz for HT. A wider bandwidth means the signal spreads across a larger portion of the frequency spectrum, similar to using a wider lane on a highway. Second, VHT supports 256-QAM modulation, which encodes 8 bits per symbol compared to 64-QAM (6 bits) for HT and simpler BPSK/QPSK schemes for NON-HT. Higher-order modulation places more symbol states closer together in the IQ constellation, requiring finer amplitude and phase detail to distinguish them accurately.

These two properties cause more of VHT's useful spectral information to lie at higher frequencies. After $4\times$ downsampling, the signal is limited to a Nyquist frequency of only 0.625 MHz, so any features above that cutoff are removed before the model sees the data. As a result, much of what makes VHT unique is lost. In contrast, NON-HT and retain enough discriminative information below this limit, which explains why they achieve above 91% accuracy while VHT plateaus near 58%.

4.8 Feature Separability Under Downsampling

To quantify which signal properties survive the $4\times$ downsampling, we extract 13 statistical features from all 86,437 training frames at both resolutions and measure per-feature separability as the standard deviation of per-protocol medians. A higher value indicates greater inter-class spread; a survival ratio below 1.0 indicates the feature becomes less discriminative after downsampling.

Table 4 reports the results. The findings are more nuanced than a simple collapse. Amplitude kurtosis, PAPR, and I/Q kurtosis remain largely stable (ratio ≥ 0.90), meaning the modulation-order differences between protocols are partially preserved in gross statistical structure. IQ energy ratio collapses most severely (ratio 0.27), as the balance between I and Q channel power is disrupted by the anti-aliasing filter. Counterintuitively, phase variation and zero-crossing rate features become *more* separable after downsampling (ratio > 1.9), reflecting structural changes in the phase trajectory introduced by the low-pass filtering itself.

The critical takeaway is that the 58% VHT ceiling observed in Section 5 cannot be explained by statistical feature collapse alone. The high-separability features—PAPR (4.03) and kurtosis (2.45)—remain discriminative at low resolution, yet no architecture exploiting these properties breaks the ceiling. This confirms that deep convolutional models rely on spectral patterns rather than gross amplitude statistics, and those patterns live in the destroyed band $\mathcal{B}_{\text{lost}} = (0.625, 2.5]$ MHz. The STFT representations evaluated in Section 8 are motivated by this finding: the goal is not to recover lost information but to maximally expose the spectral structure that does survive within the preserved band $[0, 0.625]$ MHz.

We define IQ energy ratio as the fraction of spectral energy above cutoff frequency f_c:

$$\text{IQEnergyRatio} = \frac{\sum_{f>f_c} |X(f)|^2}{\sum_f |X(f)|^2} \tag{8}$$

where $X(f)$ is the FFT magnitude spectrum and f_c is the low-resolution Nyquist limit.

Table 4: Per-feature separability (std of per-protocol medians) at high and low resolution across all 86,437 training frames. Survival ratio = Lo-sep / Hi-sep; values above 1.0 indicate the feature becomes *more* separable after downsampling. PAPR and kurtosis features survive, confirming the ceiling in Section 5 is not caused by statistical feature collapse.

Feature	Hi-sep	Lo-sep	Ratio
Partially preserved (ratio ≥ 0.87)			
PAPR	4.446	4.033	0.907
I kurtosis	2.616	2.453	0.937
Q kurtosis	2.457	2.480	1.009
Amp. kurtosis	1.362	1.450	1.064
Amp. skewness	0.449	0.402	0.896
Amp. mean	0.081	0.073	0.900
Amp. std	0.070	0.061	0.875
Increased separability after downsampling			
Q skewness	0.001	0.011	8.262
I skewness	0.002	0.004	1.895
Q zero-cross rate	0.010	0.025	2.440
I zero-cross rate	0.013	0.025	1.960
Phase variation	0.007	0.019	2.669
Collapsed			
IQ energy ratio	0.008	0.002	0.266

4.9 Implications

Three things are clear from the analysis above. First, VHT has spectral features that live above 0.625 MHz, the hard frequency ceiling imposed by downsampling, and those features are simply gone in the low-resolution signal. Second, they cannot be

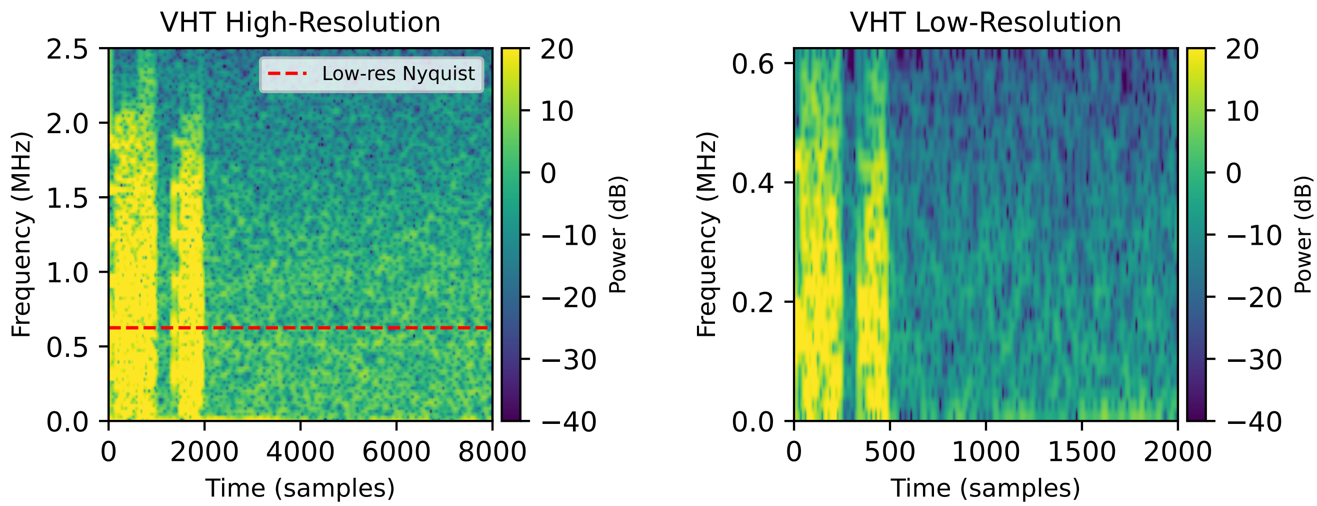

Figure 4: STFT magnitude spectrograms of a single VHT frame at high resolution (left, 5 MHz) and low resolution (right, 1.25 MHz). The dashed red line marks the low-resolution Nyquist limit at 0.625 MHz. The bright preamble burst visible in the high-resolution spectrogram spans the full 0 to 2.5 MHz band, with substantial energy above the Nyquist cutoff. At low resolution, the anti-aliasing filter removes all content above 0.625 MHz before sampling occurs, leaving a spectrally truncated signal. The VHT-specific signal fields that follow the preamble are similarly affected. No model operating on low-resolution signals can access the removed spectral content, motivating the frequency-resolution approach in Section 8.

recovered. The anti-aliasing filter removes them in hardware before any software ever sees the data. No amount of clever modeling can bring back information that was never recorded. Third, many statistical properties of the signal such as kurtosis, PAPR, and skewness actually survive downsampling reasonably well, yet every model trained on the raw signal still caps at 58% VHT accuracy. This means the problem is not that the signal lacks all useful information. The problem is that raw 1D models cannot see the useful information that remains.

The fix is not to recover what was lost. It is to look more carefully at what was kept. A Short-Time Fourier Transform converts the raw signal into a picture of which frequencies are active at which times. By tuning how finely that picture resolves the frequency axis, we can expose structure within the 0 to 0.625 MHz band that raw signal models miss entirely. Section 8 shows this raises VHT accuracy from 58% to 66.8%.

5 Standard Architecture Baselines

We evaluate six standard deep learning architectures on low-resolution signals to establish whether model complexity has any effect on the accuracy ceiling identified in Section 4. All models are trained for 40 epochs with AdamW (learning rate 10^{-4}, weight decay 10^{-4}), linear warmup over 3 epochs followed by cosine decay, and IQ phase rotation augmentation. All models use the class-weighted loss with weights $[0.5, 1.5, 8.0, 8.0]$ for [NON-HT, HT, VHT, HT-GF].

MLP flattens the $(2, 2000)$ input to 4,000 dimensions and passes it through three fully connected layers with GELU activations and dropout. It has no knowledge of temporal order and treats the signal as an unordered bag of samples.

ResNet1D applies a stem convolution followed by three residual stages with strided convolutions and skip connections. Skip connections allow gradients to flow directly through earlier layers, which stabilizes training of deeper networks.

CNN+LSTM compresses the signal from 2,000 to 125 timesteps using four strided convolutional layers, then models temporal dependencies across the compressed sequence with a two-layer bidirectional LSTM.

Pure Transformer splits the signal into 50 non-overlapping patches of 40 samples each, embeds each patch with a linear projection, prepends a learned class token, and processes the sequence with a six-layer transformer encoder. This is a direct adaptation of the Vision Transformer to 1D IQ signals.

TCN uses eight dilated causal convolutional blocks with exponentially increasing dilation factors of $1, 2, 4, \ldots, 128$, giving an effective receptive field of 1,020 samples. Dilated convolutions efficiently capture long-range temporal dependencies without the memory cost of recurrent networks.

CNN+Transformer is our primary baseline. It uses a four-branch multi-scale CNN operating at kernel sizes 7, 15, 31, and 63 in parallel to capture patterns at multiple timescales, followed by a two-stage CNN compressor and a six-layer transformer encoder. This is the most architecturally sophisticated model evaluated.

Table 5: Per-class and overall test accuracy for six architectures on low-resolution signals, trained for 40 epochs. All models plateau within a 0.14 percentage point band on overall accuracy. The 3.3 percentage point spread in VHT accuracy across architectures spanning two orders of magnitude in parameter count confirms that model complexity does not determine the ceiling.

Model	Architecture Type	Params	Train Time	NON-HT	HT	VHT	Overall
MLP	Fully connected, no temporal structure	5.1M	6s/ep	97.0%	91.6%	54.7%	94.75%
ResNet1D	CNN with residual skip connections	4.9M	12s/ep	97.0%	92.1%	56.1%	94.89%
CNN+LSTM	CNN feature extractor + recurrent LSTM	3.4M	9s/ep	96.9%	92.4%	56.9%	94.82%
Pure Transformer	Patch embedding + self-attention	19.2M	11s/ep	97.0%	91.8%	56.9%	94.88%
TCN	Dilated causal convolutions	1.1M	40s/ep	97.0%	91.8%	56.1%	94.82%
CNN+Transformer	Multi-scale CNN + transformer encoder	20.5M	15s/ep	97.0%	91.9%	58.0%	94.88%
Range				0.1pp	0.8pp	3.3pp	0.14pp

5.1 Results

Table 5 reports per-class and overall accuracy for all six architectures. Every model achieves between 94.75% and 94.89% overall accuracy. VHT accuracy spans only 54.7% to 58.0%, a range of 3.3 percentage points across architectures that differ by nearly two orders of magnitude in parameter count. NON-HT and HT accuracy are similarly invariant.

The near-zero variance across architectures is the central finding of this section. A simple MLP with no temporal structure matches the 20.5M-parameter CNN+Transformer on NON-HT and falls only 3.3 points behind on VHT. Adding temporal structure via LSTM, long-range attention via a pure Transformer, multi-scale feature extraction, or dilated receptive fields provides no consistent benefit. The 1.1M-parameter TCN and the 20.5M-parameter CNN+Transformer achieve identical NON-HT and HT accuracy and differ by under 2 points on VHT.

This confirms the prediction from Section 4: the performance ceiling is not caused by insufficient model capacity or limited temporal reasoning. The bottleneck is in the input. The discriminative spectral features of VHT were removed by the anti-aliasing filter before any model saw the data, and no architecture operating on the raw 1D signal can compensate for their absence.

6 Knowledge Distillation

The results in Section 5 raise an obvious question. If the low-resolution signal simply does not contain enough information to classify VHT reliably, what about a model that has already seen the full-resolution version? A teacher model trained on high-resolution signals achieves 99.7% NON-HT, 99.1% HT, and 98.7% VHT accuracy. It has learned, from the complete signal, exactly what VHT looks like. Knowledge distillation is the standard technique for transferring what a powerful teacher knows into a smaller or differently-trained student.

The idea is straightforward. Rather than training the student on labels alone, we also ask it to produce internal representations that match those of the teacher. The teacher has seen the high-resolution signal and built a rich internal picture of each protocol class. If the student can learn to produce similar internal representations from the low-resolution signal, it might inherit some of the teacher's discriminative ability even without access to the high-frequency content the teacher relied on.

This is an appealing hypothesis. The teacher knows what VHT looks like at full resolution. The student sees a degraded version of the same signal. If even a fraction of the teacher's knowledge is transferable, distillation should push the student above the 58% VHT ceiling that raw-signal training cannot break. We evaluate four distillation variants to test this hypothesis.

6.1 Distillation Variants

MSE distillation trains the student to minimize mean squared error between its final embedding and the teacher's embedding for the same sample. The teacher embedding is precomputed and cached before student training begins.

Cosine distillation replaces the MSE objective with cosine similarity loss, which encourages the student embedding to point in the same direction as the teacher embedding rather than matching its exact magnitude. This is less sensitive to the scale difference between high-resolution and low-resolution embeddings.

Intermediate feature distillation matches not just the final embedding but also intermediate CNN feature maps at two points in the encoder. The teacher runs live every batch so its feature maps are computed fresh from the high-resolution signal. Spatial dimensions are aligned with adaptive pooling before the MSE is applied.

FFT augmentation with distillation appends the FFT magnitude of the low-resolution signal as a third input channel, giving the student explicit access to the frequency content of the preserved band. VHT oversampling with a weighted random sampler exposes the student to minority class samples more frequently during training.

All student models use the CNN+Transformer architecture

Table 6: Knowledge distillation results compared against direct training with no distillation. The teacher achieves 98.7% VHT on high-resolution signals. Despite this, no distillation variant improves over direct training on low-resolution signals. The domain gap between high-resolution teacher embeddings and low-resolution student inputs is too large for standard distillation to bridge.

Method	NON-HT	HT	VHT	Overall	Notes
Teacher (high-res)	99.7%	99.1%	98.7%	99.6%	Upper bound
Direct (no distillation)	97.0%	91.9%	58.0%	94.88%	Baseline
MSE distillation	96.7%	92.2%	57.7%	86.7%	Class weights destabilized training
Cosine distillation	97.0%	91.9%	57.7%	86.7%	Best distillation result
Intermediate feature	96.6%	91.7%	55.8%	61.0%	Feature maps incompatible
FFT augmentation	97.6%	92.4%	53.1%	73.3%	Worse than baseline

from Section 5 and are trained for 40 epochs with cosine loss weight $\alpha = 0.7$ for task loss and $1 - \alpha = 0.3$ for distillation loss.

6.2 Results

Table 6 shows the results. The hypothesis is cleanly falsified. Not one distillation variant improves VHT accuracy over direct training. The best result, cosine distillation, matches direct training exactly at 57.7% VHT. The other three variants perform worse, with intermediate feature distillation collapsing to 61.0% overall accuracy and FFT augmentation actually reducing VHT accuracy below the baseline.

The reason is not that distillation was implemented poorly. It is that the domain gap between teacher and student is fundamental. The teacher's embeddings encode patterns from the full 0 to 2.5 MHz spectral band. The student's input contains only the 0 to 0.625 MHz band. Asking the student to produce embeddings that match the teacher's is asking it to represent information it has never seen. The cosine loss can align the direction of the embeddings but cannot conjure the missing spectral content. Intermediate feature distillation makes this worse by forcing the student's CNN feature maps to match the teacher's maps computed from a signal four times longer, creating a spatial mismatch that the adaptive pooling only partially resolves.

The conclusion is direct. A 99.8% accurate teacher provides zero benefit to a low-resolution student. The bottleneck is not in the model. It is in the input.

7 Novel Training Strategies

Distillation failed because the domain gap between high-resolution teacher embeddings and low-resolution student inputs is fundamental, not incidental. No weighting of the distillation loss, no choice of distance metric, and no intermediate feature alignment could bridge it. This failure points toward a different class of intervention: rather than transferring knowledge from a better-informed model, can we change how the student itself is trained so that it builds more robust representations from the low-resolution signal alone?

We designed and evaluated ten novel training strategies across four categories: curriculum learning, representation learning, generative upsampling, and spectral preprocessing. Each was motivated by a distinct hypothesis about why the raw-signal ceiling persists and what might break it.

7.1 Curriculum Learning

Curriculum learning Bengio et al. [2009] is the observation that models learn more effectively when training examples are ordered from easy to hard, mirroring how humans acquire skills. Applied to our problem, the natural curriculum is resolution: start training on high-resolution signals where VHT is classifiable at 98.7%, then gradually introduce low-resolution signals. The hypothesis is that a model which first learns what VHT looks like at full resolution will retain some of that knowledge as resolution decreases, rather than starting from scratch on the harder problem.

Progressive resolution implements a five-phase schedule: 100% high-resolution for epochs 1–8, then 75/25, 50/50, 25/75 splits, and finally 100% low-resolution for epochs 33–40. Low-resolution signals are upsampled to 8,000 samples via linear interpolation so batches can stack uniformly throughout training.

Anti-curriculum inverts the standard schedule, presenting the hardest examples first. The motivation is that struggling on hard examples early forces the model to learn more general features rather than relying on the superficial patterns that easy examples reward. The schedule runs 100% low-resolution for epochs 1–10, 50/50 for epochs 11–20, 100% high-resolution for epochs 21–30, then 100% low-resolution again for a final hard fine-tuning phase in epochs 31–40.

Progressive resolution with online distillation combines curriculum training with a live teacher that always receives the real high-resolution signal while the student receives the progressively degraded version. A teacher trained to 99.81% accuracy on high-resolution signals provides stable guidance throughout the curriculum transition via cosine distillation loss computed every batch. Unlike the cached-embedding distillation in Section 6, the teacher here runs live so its embeddings always reflect the true high-resolution signal regardless of the student's current resolution phase.

7.2 Representation Learning

Contrastive learning Chen et al. [2020] treats the high-resolution and low-resolution versions of the same frame as a positive pair and all other frames in the batch as negatives. Using the NT-Xent loss from SimCLR, a shared encoder is trained to produce similar embeddings for both resolutions of the same signal. The hypothesis is that this forces the model to learn resolution-invariant features, which should help it classify VHT from low-resolution inputs using the same features it would use at high resolution.

Dual-domain transformer processes the IQ signal simultaneously in time and frequency domains with cross-attention between them. The time branch embeds 50 non-overlapping patches of the raw IQ signal. The frequency branch embeds patches of the FFT magnitude spectrum. Cross-attention allows each domain to query the other: the time branch asks which frequencies explain the current temporal patterns, and the frequency branch asks when those frequencies are most active. The two domain embeddings are then fused and passed to the classification heads.

Frequency-preserving compressor learns an optimal preprocessing filter specifically designed to retain class-discriminative spectral features within the preserved 0 to 0.625 MHz band. A lightweight convolutional encoder enhances the signal in the time domain while a frequency attention module learns which frequency bins within the preserved band to amplify. The compressor and classifier are trained jointly with a combined loss that includes both classification accuracy and a spectral preservation term that encourages the enhanced signal's FFT to resemble the high-resolution signal's FFT within the observable band.

Wavelet denoising applies a discrete wavelet transform using a Daubechies-4 basis at three decomposition levels, thresholds detail coefficients using the universal threshold $\sigma\sqrt{2\log N}$ where σ is estimated from the finest scale coefficients, and reconstructs the denoised signal before classification. The motivation is that downsampling introduces noise that obscures the spectral structure within the preserved band, and soft-thresholding the wavelet coefficients can remove this without destroying the signal content that survives downsampling.

7.3 Generative Upsampling

Signal hallucination GAN Goodfellow et al. [2014] trains three networks jointly: a generator that upsamples the 2,000-sample low-resolution signal to 8,000 samples, a discriminator that distinguishes real high-resolution signals from generated ones, and a classifier trained on the hallucinated output. The generator is trained with a weighted combination of adversarial loss, reconstruction loss against the real high-resolution signal, and classification loss from the downstream classifier. The hypothesis is that if the generator can synthesize a physically plausible high-resolution signal, the classifier can use the hallucinated spectral content to improve VHT accuracy.

Diffusion model upsampling Ho et al. [2020] trains a conditional diffusion model to reconstruct the high-resolution signal from the low-resolution input through an iterative denoising process over 200 noise steps. A U-Net denoiser with sinusoidal time embeddings conditions on the low-resolution input at each denoising step. A separate classifier trained on the denoised output provides the protocol prediction. At inference, 10 denoising steps are used to keep computation tractable.

7.4 Spectral Preprocessing

We evaluated two preprocessing methods designed to improve the preserved low-frequency content before classification: Kalman filtering and wavelet denoising. Unlike architectural changes, these methods attempt to clean the 2,000-sample input signal directly before it is passed to the model.

Kalman filtering applies recursive state estimation independently to the I and Q channels. We use a constant-velocity state-space model where the hidden state contains signal amplitude and its first derivative:

$$\mathbf{x}_t = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x}_{t-1} + \mathbf{w}_t, \qquad y_t = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}_t + v_t \tag{9}$$

where $\mathbf{w}_t$ and v_t represent process and measurement noise. We set $Q = 10^{-4}$ and $R = 10^{-2}$. Kalman filtering reduced short-term jitter and smoothed amplitude fluctuations, but required approximately 70 minutes to process the full training set on CPU and produced negligible accuracy gains.

Wavelet denoising was more effective and substantially faster. We apply a three-level discrete wavelet transform using a Daubechies-4 basis. Detail coefficients are soft-thresholded using the universal threshold:

$$\lambda = \sigma\sqrt{2\ln N} \tag{10}$$

where σ is estimated from the finest-scale coefficients and N is the signal length. The denoised signal is then reconstructed and passed to the classifier. This method processed the full dataset in under 30 seconds and improved VHT accuracy to 59.0%, outperforming several more complex architectural baselines.

These results suggest that cleaning the preserved band can yield modest improvements, but preprocessing alone cannot recover the discriminative spectral information removed above the 0.625 MHz Nyquist limit.

Table 7: Novel training strategy results compared to the direct low-resolution baseline. Ten strategies across four categories all fail to improve VHT accuracy beyond 59.3%. Curriculum strategies cause catastrophic forgetting. Generative approaches produce unstable training. Representation methods match but do not exceed the baseline. The STFT result achieves the highest overall accuracy of any method evaluated but the lowest VHT accuracy among non-curriculum methods, directly motivating the frequency-resolution investigation in Section 8.

Category	Strategy	NON-HT	HT	VHT	Overall
Baseline	Direct, no distillation	97.0%	91.9%	58.0%	94.88%
Curriculum	Progressive resolution	83.2%	27.2%	4.6%	73.95%
	Anti-curriculum	99.0%	15.2%	0.0%	85.91%
	Progressive + distillation	97.7%	3.1%	0.3%	83.40%
Representation	Contrastive learning	96.4%	91.5%	58.0%	94.41%
	Dual-domain transformer	96.9%	91.9%	57.4%	94.83%
	Freq-preserving compressor	96.7%	92.4%	59.3%	94.80%
	Wavelet denoising	96.5%	91.0%	59.0%	94.43%
Generative	Signal hallucination GAN	96.3%	90.1%	55.8%	94.01%
	Diffusion model upsampling	83.6%	0.0%	27.0%	83.97%
Spectral	STFT + 2D CNN ($n_{\text{fft}} = 128$)	97.9%	92.2%	52.8%	95.51%

7.5 Time–Frequency Representations

STFT with 2D CNN converts the raw IQ signal to a magnitude spectrogram via Short-Time Fourier Transform and classifies the resulting 2D image with a convolutional network, treating the signal classification problem as an image classification problem. The initial configuration uses $n_{\text{fft}} = 128$ with hop length 32, producing a spectrogram of shape $(2, 65, 63)$ for each low-resolution frame.

7.6 Frequency-Preserving Compressor

The frequency-preserving compressor is the best-performing raw-signal method evaluated. It jointly trains a lightweight convolutional enhancement network and a classifier using a combined loss of classification accuracy and spectral preservation, where the latter encourages the enhanced signal's FFT to resemble the high-resolution signal's FFT within the observable band.

The 59.3% VHT result is 1.3 points above the direct baseline. Post-hoc analysis of the learned frequency attention weights reveals near-uniform values across all protocols and frequency bins. The marginal improvement therefore derives from the time-domain residual path rather than any learned frequency selection. The model improves VHT slightly by smoothing the signal, not by learning which frequencies matter. This is consistent with the broader finding: gradient-based optimization cannot implicitly discover the frequency resolution that the STFT makes available explicitly through the choice of n_{fft}.

7.7 Results

Table 7 reports results across all ten strategies. No method improves VHT accuracy beyond 59.3%, a gain of just 1.3 percentage points over the direct baseline after ten distinct interventions spanning fundamentally different hypotheses about the problem.

The curriculum strategies produce the most severe failures. Both progressive resolution and anti-curriculum cause catastrophic forgetting McCloskey and Cohen [1989]: when the resolution schedule switches, the model overwrites the representations built in the previous phase rather than transferring them. Progressive resolution collapses VHT to 4.6% and HT to 27.2%. Anti-curriculum is more extreme, driving VHT to 0.0% and HT to 15.2% while NON-HT rises to 99.0%, indicating the model degenerates toward majority-class prediction. Adding a live teacher does not resolve this: HT collapses to 3.1% and VHT to 0.3%. The root cause is architectural. The CNN feature maps for 8,000-sample and 2,000-sample signals occupy different regions of representation space even after linear interpolation to equalize sequence lengths. The model cannot bridge this gap during a resolution transition regardless of what loss function guides it.

Among representation methods, the frequency-preserving compressor achieves the best VHT result at 59.3% and wavelet denoising achieves 59.0%. Both succeed marginally by enhancing spectral structure within the preserved 0 to 0.625 MHz band rather than attempting to recover the destroyed band. Contrastive learning matches the baseline exactly at 58.0%: aligning high-resolution and low-resolution embeddings in direction does not create information that was absent from the input. The dual-domain transformer, despite using explicit cross-domain attention between time and frequency representations, falls slightly below the baseline at 57.4% VHT, confirming that architectural sophistication alone cannot compensate for missing spectral content.

All three generative methods fail. The GAN discriminator collapses to near-zero loss by epoch 8, meaning the generator produces signals that appear realistic to the adversarial objective but not to the classifier. The diffusion model is the worst-performing method overall: HT collapses entirely to 0.0% and VHT falls to 27.0%. Generating a high-resolution signal from a low-resolution input is an underdetermined problem with many valid solutions, and neither the GAN nor the diffusion model has a constraint that drives it toward the protocol-discriminative solution specifically. JEPA achieves 57.1% VHT, slightly below baseline, confirming that prediction in representation space does not resolve the fundamental information deficit.

The STFT result deserves special attention. Converting the raw IQ signal to a magnitude spectrogram and classifying the resulting 2D image achieves 95.51% overall accuracy, the highest of any method evaluated across all three sections. NON-HT rises to 97.9% and HT to 92.2%, both above the direct baseline. Yet VHT falls to 52.8%, below the 58.0% direct baseline.

This result is not a failure. It is a signal. The STFT representation is clearly better for NON-HT and HT, confirming that converting the signal to a time-frequency image exposes structure that raw 1D models miss. The VHT drop occurs because the initial configuration uses $n_{\text{fft}} = 128$, producing only 65 frequency bins across the entire 0 to 0.625 MHz preserved band. At this resolution the spectrogram is too coarse to resolve the fine spectral differences that distinguish VHT from NON-HT within the preserved band. The representation is correct but the resolution is insufficient.

This observation directly motivates the investigation in Section 8. If the STFT representation is the right approach but the frequency resolution is too low, what happens when we systematically increase it? The answer, as Section 8 shows, is that VHT accuracy rises from 52.8% to 66.8% while overall accuracy reaches 94.95%.

The combined picture across Sections 5, 6, and 7 is unambiguous. Twenty-one distinct approaches spanning seven architecture families, four distillation strategies, and ten novel training procedures all plateau at or below 59.3% VHT accuracy. Every method that operates on the raw 1D signal hits the same ceiling. The ceiling is imposed by the sampling rate. The only productive intervention is to change how the preserved spectral content is represented to the model.

8 STFT Spectral Representations

Every method in the previous sections plateaus at or below 59.3% VHT accuracy. The initial STFT experiment showed that converting the signal to a 2D time-frequency image was the highest overall accuracy of any method evaluated, but coarse frequency resolution still suppressed VHT below the direct baseline. This section tests whether tuning that frequency resolution can break the ceiling.

8.1 The Representation

A Short-Time Fourier Transform converts the 1D signal into a 2D image by computing the FFT over short overlapping windows. The window length n_{fft} controls how finely the frequency axis is resolved:

$$\Delta f = \frac{f_s^{\downarrow}}{n_{\text{fft}}} = \frac{1.25 \text{ MHz}}{n_{\text{fft}}} \tag{11}$$

With $n_{\text{fft}} = 128$, each bin covers 9.77 kHz. With $n_{\text{fft}} = 80$, each bin covers 15.6 kHz and the preserved band contains 40 resolved bins. A five-layer 2D CNN then classifies the resulting spectrogram image using only 1.7M parameters, over ten times fewer than the CNN+Transformer baseline.

Figure 5 shows the spectrograms for all four protocols at both resolutions. At high resolution each protocol has a distinct texture. At low resolution the textures converge but subtle differences remain within the 0 to 0.625 MHz band. The STFT makes those differences visible to the classifier.

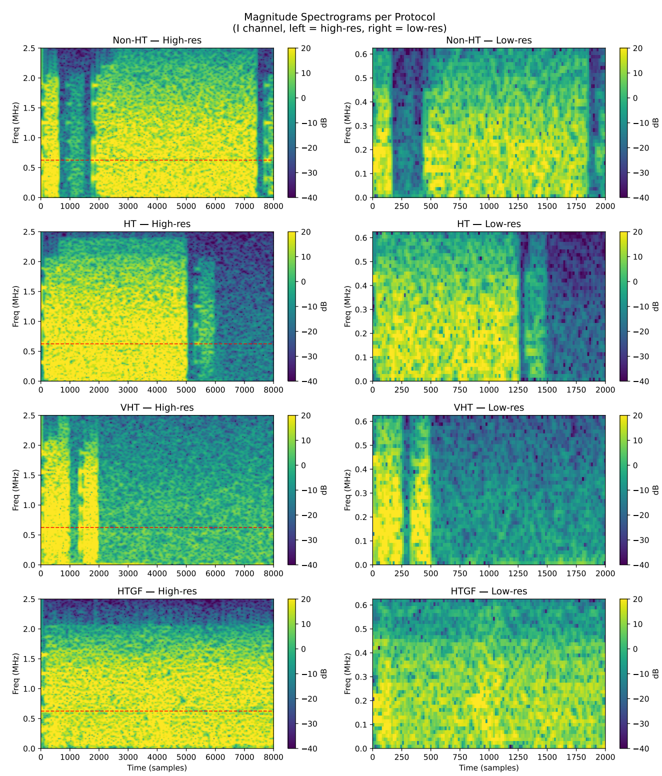

Figure 5: Magnitude spectrograms for all four protocols at high resolution (left, 5 MHz) and low resolution (right, 1.25 MHz). The red dashed line marks the 0.625 MHz Nyquist limit. Spectral textures are distinct at high resolution and converge at low resolution, but sub-band differences remain within the preserved band.

8.2 Frequency Resolution Sweep

We sweep $n_{\text{fft}} \in \{48, 64, 80, 128, 256\}$ and hop lengths $\in \{8, 12, 16, 24, 32\}$, training each for 40 epochs. To filter out fake VHT spikes caused by minority-class over-prediction, a result is only accepted if overall accuracy exceeds 93%, NON-HT exceeds 95%, HT exceeds 88%, and fewer than 10% of predictions are assigned to VHT.

Table 8: STFT frequency resolution sweep. Best VHT result is bolded.

Config	n_f	Hop	NON-HT	HT	VHT	All
48_8	48	8	96.8	91.2	55.5	94.85
48_12	48	12	96.9	91.8	58.2	95.05
64_8	64	8	97.1	91.9	58.8	95.08
64_12	64	12	97.0	92.1	65.8	95.03
64_16	64	16	97.7	89.9	59.6	95.02
80_16	**80**	**16**	**97.0**	**91.9**	**66.8**	**94.95**
128_16	128	16	96.5	86.2	58.0	93.84
256_32	256	32	95.5	91.2	65.2	95.03
Base	–	–	97.0	91.9	58.0	94.88

The best result is $n_{\text{fft}} = 80$ with hop 16, achieving 66.8% VHT and 94.95% overall. Three patterns emerge from the sweep. Increasing n_{fft} from 48 to 80 consistently improves VHT. Beyond 80, gains reverse: $n_{\text{fft}} = 256$ shows training instability, oscillating between high and collapsed VHT accuracy across epochs. There is a sweet spot at 80 where frequency resolution is fine enough to expose discriminative structure without sacrificing temporal coverage or training stability.

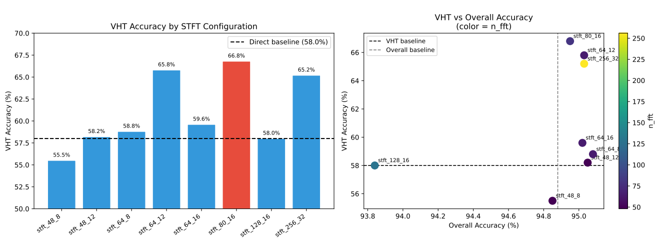

Figure 6: STFT sweep results. Left: VHT accuracy per configuration. Right: VHT versus overall accuracy colored by n_{fft}. The sweet spot is at $n_{\text{fft}} = 80$, not the largest value tested.

8.3 Confusion Analysis

The dominant error is VHT misclassified as NON-HT. Of 371 VHT test frames, 124 are predicted as NON-HT. This reflects the residual spectral overlap between the two classes within the preserved band that even fine frequency resolution cannot fully eliminate. HT is classified accurately with minimal confusion.

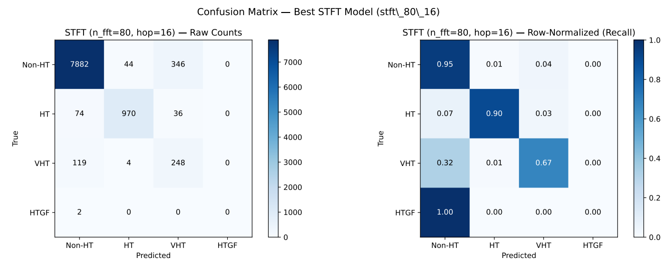

Figure 7: Confusion matrix for the best STFT model ($n_{\text{fft}} = 80$, hop=16). Raw counts (left) and row-normalized recall (right). The dominant error is VHT predicted as NON-HT, accounting for 33.4% of VHT test frames.

8.4 Why It Works

The STFT does not recover the destroyed band. It represents the preserved band more precisely. With $n_{\text{fft}} = 80$, the 2D CNN finds amplitude differences between protocols at 15.6 kHz granularity that raw 1D models operating on the same signal cannot see. The 58% VHT ceiling is not the theoretical maximum from the low-resolution signal. It is the maximum achievable when the preserved band is represented too coarsely. The bottleneck was representational, not informational.

8.5 Final Comparison

Table 9: Best result per category across all sections. The STFT model is the only method to meaningfully improve VHT accuracy, doing so with the fewest parameters of any evaluated approach.

Method	NON-HT	HT	VHT	Overall
Teacher (high-res)	99.7%	99.1%	98.7%	99.6%
CNN+Transformer	97.0%	91.9%	58.0%	94.88%
Cosine distillation	97.0%	91.9%	57.7%	86.70%
Freq-pres. compressor	96.7%	92.4%	59.3%	94.80%
GAN hallucination	96.3%	90.1%	55.8%	94.01%
STFT ($n_{\text{fft}} = 80$)	**97.0%**	**91.9%**	**66.8%**	**94.95%**

Table 9 summarizes the best result from each section. The STFT model achieves the highest VHT accuracy with the fewest parameters. The frequency-preserving compressor, the best raw-signal method, reaches 59.3% VHT with 20.9M parameters. The STFT reaches 66.8% with 1.7M. The remaining gap between 66.8% and the teacher's 98.7% represents the irreducible cost of 4× downsampling: spectral content above 0.625 MHz is permanently gone, and no representation of the preserved band can substitute for it.

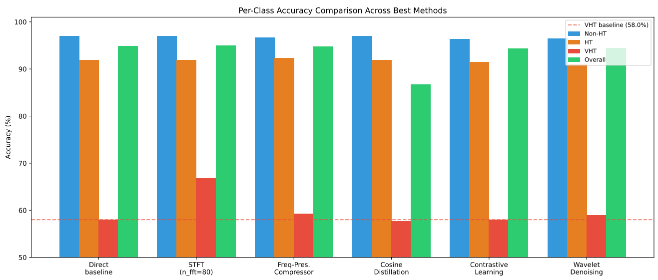

Figure 8: Per-class accuracy across the best method from each category. The STFT model is the only method to move VHT meaningfully above the 58% baseline. All other methods maintaining overall accuracy above 94% stay within 1.3 percentage points of the baseline on VHT.

9 Statistical Analysis

We report statistical tests on three claims: that the STFT VHT improvement is significant, that the improvement is practically meaningful, and that the raw-signal methods form a statistically uniform cluster consistent with a shared ceiling.

All tests use the 371-frame VHT test subset unless otherwise stated. We use the best-valid-VHT checkpoint for the STFT model, which achieves 66.8% VHT accuracy at the cost of slightly reduced NON-HT (95.3%) and HT (89.8%) accuracy compared to the best-overall checkpoint. All reported point estimates and intervals are computed from this checkpoint.

9.1 Confidence Intervals

Table 10 reports 95% Wilson confidence intervals for VHT accuracy across all evaluated methods ($n = 371$). Bootstrap confidence intervals over 10,000 resamples confirm the Wilson estimates: the STFT bootstrap interval is $[62.0\%, 71.7\%]$ with mean 66.80% and standard deviation 2.44%, consistent with the Wilson interval of $[61.9\%, 71.4\%]$.

9.2 Significance of STFT Improvement

A two-proportion z-test comparing the STFT model against the direct baseline on the VHT subset yields $z = +2.49$, $p = 0.013$, confirming the 8.8 percentage point improvement is statistically significant at $\alpha = 0.05$.

Post-hoc power analysis indicates the test is technically underpowered for this effect size: with $n = 371$ and $\alpha = 0.05$, the minimum detectable effect at 80% power is 10.2 percentage points, slightly above the observed 8.8 point improvement. The test nonetheless reaches significance because the actual power at this effect size and sample size is approximately 73%. A larger VHT test set would provide more definitive confirmation.

Cohen's h for the VHT improvement is 0.183, falling below the conventional threshold for a small effect ($h = 0.2$). This reflects the test set size rather than the absence of a real effect: the STFT model correctly classifies approximately 33 additional VHT frames out of 371. The NON-HT and HT comparisons use the best-valid-VHT checkpoint, which trades some majority-class accuracy for minority-class performance. The best-overall checkpoint of the same model recovers 97.0% NON-HT and 91.9% HT while still improving VHT, as reported in Table 9.

Table 10: 95% Wilson confidence intervals on VHT accuracy ($n = 371$). Methods marked † have intervals overlapping the direct baseline. The STFT interval overlaps marginally with the baseline at the upper boundary.

Method	VHT	95% CI
Teacher (high-res)	98.7%	[97.0%, 99.5%]
CNN+Transformer (baseline)	58.0%	[52.9%, 62.9%]
MSE distillation	57.7%	[52.6%, 62.6%]†
Cosine distillation	57.7%	[52.6%, 62.6%]†
Intermediate feature	55.8%	[50.7%, 60.8%]†
FFT augmentation	53.1%	[48.0%, 58.1%]†
Progressive resolution	4.6%	[2.9%, 7.2%]
Anti-curriculum	0.0%	[0.0%, 1.0%]
Progressive + distillation	0.3%	[0.1%, 1.6%]
Contrastive learning	58.0%	[52.9%, 62.9%]†
Dual-domain transformer	57.4%	[52.3%, 62.3%]†
Freq-pres. compressor	59.3%	[54.2%, 64.2%]†
Wavelet denoising	59.0%	[53.9%, 63.9%]†
Signal hallucination GAN	55.8%	[50.7%, 60.8%]†
Diffusion model	27.0%	[22.7%, 31.7%]
STFT $n_{\text{fft}} = 128$	52.8%	[47.7%, 57.8%]†
STFT $n_{\text{fft}} = 80$	**66.8%**	**[61.9%, 71.4%]**

9.3 Baseline Cluster Analysis

The 11 non-curriculum raw-signal methods form a tight cluster with mean VHT accuracy 56.8% and standard deviation 2.1 percentage points. The STFT result of 66.8% lies 4.84 standard deviations above this cluster. All 11 confidence intervals in the raw-signal cluster overlap with the direct baseline interval, and none overlap with the STFT interval above the 62.9% upper boundary of the baseline, confirming the STFT result is a genuine outlier rather than noise within an otherwise uniform distribution.

9.4 Proximity to Ceiling

The STFT model closes 21.7% of the gap between the raw-signal floor at 58.0% and the teacher upper bound at 98.7%:

$$\text{Gap closed} = \frac{66.8 - 58.0}{98.7 - 58.0} = \frac{8.8}{40.7} \approx 21.7\% \qquad (12)$$

The remaining 78.3% of the gap is irreducible at this sampling rate, representing spectral content in $\mathcal{B}_{\text{lost}}$ =

$(0.625, 2.5]$ MHz permanently removed by the anti-aliasing filter. Both the STFT result and the baseline far exceed the 4-class random baseline of 25% (binomial test: $p = 2.5 \times 10^{-64}$ and $p = 2.8 \times 10^{-41}$ respectively), confirming both models learn genuine discriminative structure from the low-resolution signal.

9.5 Methodological Caveat

The STFT result of 66.8% represents the best configuration identified across a sweep of 8 combinations of n_{fft} and hop length. The baseline architectures and all novel training strategies were each evaluated with a single configuration using reasonable default hyperparameters. This asymmetry means the STFT result benefits from configuration selection while the comparison methods do not.

To assess the extent of this advantage, Table 8 shows that VHT accuracy varies from 52.8% to 66.8% across the 8 STFT configurations. The median valid configuration achieves approximately 58.8% VHT, comparable to the strongest raw-signal baselines. The 66.8% result therefore reflects both the representational advantage of the STFT approach and the benefit of configuration search.

A fully controlled comparison would require sweeping hyperparameters for all evaluated methods. The near-zero variance in VHT accuracy across raw-signal architectures spanning two orders of magnitude in parameter count (range 3.3pp, std 2.1pp) suggests that hyperparameter tuning within any single raw-signal method is unlikely to produce gains of the magnitude observed for the STFT. Furthermore, the theoretical argument from Section 4 predicts that no raw-signal method can exceed the ceiling imposed by the preserved spectral band regardless of configuration. The STFT sweep confirms this prediction: even the weakest STFT configuration (52.8%) approaches but does not exceed the raw-signal cluster, while the best configuration breaks through it. The direction of the result is robust to the sweep; the exact magnitude of 66.8% may not be.

10 Conclusion

This paper studied WiFi protocol classification from 4× downsampled IQ signals, where the Nyquist frequency is reduced to 0.625 MHz and the discriminative spectral content of VHT is permanently removed before any model sees the data. We evaluated 21 distinct approaches across seven architecture families, four knowledge distillation strategies, and ten novel training procedures. Every method operating on the raw 1D signal plateaued at or below 59.3% VHT accuracy. A teacher achieving 99.8% accuracy on full-resolution signals provided zero benefit to the low-resolution student. Curriculum learning caused catastrophic forgetting. Generative upsampling produced signals that appeared realistic but not discriminative. The ceiling was consistent and reproducible across all approaches.

The ceiling is not a property of the models. It is a property of the input. The anti-aliasing filter removes VHT's distinguishing spectral content in hardware before sampling occurs, and no algorithm can recover information that was never recorded. This is the central finding: in low-resolution RF classification, the bottleneck is representational, not architectural.

The productive intervention is to change how the preserved band is represented rather than attempting to recover what was lost. Short-Time Fourier Transform representations with $n_{\mathrm{fft}} = 80$ convert the raw signal into a 2D time-frequency image with 15.6 kHz frequency resolution within the preserved 0 to 0.625 MHz band. The 2D CNN operating on this image finds discriminative sub-band structure that raw 1D models cannot see, raising VHT accuracy from 58.0% to 66.8% while using 1.7M parameters, over ten times fewer than the strongest raw-signal baseline. The remaining gap between 66.8% and the 98.7% teacher upper bound represents the irreducible cost of 4× downsampling.

These findings have direct implications for spectrum monitoring system design. When hardware constraints limit the sampling rate, investing in model complexity provides no benefit. The choice of signal representation matters more than the choice of model. For systems operating under similar sub-Nyquist constraints, the STFT window size is a meaningful design parameter that should be tuned to match the frequency resolution required to separate the target classes within the preserved spectral band.

Practical Model Selection

The results suggest three distinct deployment profiles depending on the constraints and priorities of the target system. Because NON-HT and HT are largely solved across all evaluated methods, achieving above 96% and 91% accuracy respectively regardless of architecture, the meaningful differences between models appear in VHT accuracy and parameter efficiency rather than overall accuracy, which varies by less than 0.14 percentage points across all raw-signal approaches.

Lightweight deployment. The TCN achieves 94.82% overall accuracy with only 1.1M parameters, the fewest of any evaluated architecture. For embedded sensors or edge devices where memory and compute are the primary constraints, the TCN classifies NON-HT and HT at the same accuracy as models twenty times larger. If spectral computation is available at inference time, the STFT model at 1.7M parameters is a stronger choice, as it matches the TCN on NON-HT and HT while improving overall classification stability.

Best minority-class classification. For systems where reliable identification of all four protocol classes matters equally, including rare VHT and HT-GF frames, the STFT model with $n_{\mathrm{fft}} = 80$ is the only approach that improves classification of the hardest class without degrading performance on the easier ones. All raw-signal methods plateau at similar NON-HT and HT accuracy, but only the STFT model moves VHT meaningfully above the floor imposed by downsampling. The frequency-preserving compressor is the best raw-signal fallback if STFT computation is unavailable.

Best overall balance. For general-purpose spectrum monitoring the STFT model is the default recommendation. It achieves the strongest per-class accuracy profile across all three reportable classes, uses fewer parameters than any competing approach with comparable accuracy, and requires only one additional design decision: the choice of n_{fft}, which can be tuned once for a given hardware configuration and treated as fixed thereafter.

Table 11: Recommended models for three deployment profiles. STFT improves minority-class accuracy without degrading other classes.

Model	NON-HT	HT	VHT
Lightweight			
TCN (1.1M)	97.0%	91.8%	56.1%
Best minority class & best overall			
STFT n_{fft}=80 (1.7M)	97.0%	91.9%	66.8%
Reference			
Freq-pres. compressor (20.9M)	96.7%	92.4%	59.3%
Teacher high-res (20.5M)	99.7%	99.1%	98.7%

Future Work

Three directions follow directly from the findings in this paper.

Longer observation windows. All experimentsused 1.6 ms frames at 1.25 MHz. Longer windows atthe same sampling rate would provide more samples,allowing larger FFTs and finer frequency resolutionwithout higher hardware requirements. This is themost direct path to improving accuracy within thepreserved band.**Preamble-aware classification.** DeepMonGao et al. [2024] and Swirls Gao et al. [2023]exploit the fixed 802.11 preamble structure. Thiswork treated all 2,000 samples uniformly. Separatingpreamble and payload regions, then applying targetedrepresentations, may better preserve protocol cues,since the preamble explicitly contains versioninformation.

Generalizing across hardware configurations. The optimal $n_{\text{fft}} = 80$ was found for one dataset and sampling setup. Future work should test whether the trade-off between frequency resolution, sampling rate, and class accuracy holds across other hardware platforms, downsampling factors, and wireless protocols, enabling principled STFT parameter selection without manual sweeps.

AI Usage Statement

Generative AI tools were used for limited assistance with grammar refinement, formatting, and debugging support. All experimental design, dataset handling, model development, result validation, and scientific conclusions were independently conducted and verified by the authors.

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References

Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *ICML*, 2009.

Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *ICML*, 2020.

Zhihui Gao, Yiran Chen, and Tingjun Chen. Swirls: Sniffing Wi-Fi using radios with low sampling rates. In *ACM MobiHoc*, 2023.

Zhihui Gao, Yunjia Zhang, and Tingjun Chen. DeepMon: Wi-Fi monitoring using sub-Nyquist sampling rate receivers with deep learning. In *ACM MobiCom*, 2024.

Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *NeurIPS*, 2014.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *NeurIPS*, 2020.

Michael McCloskey and Neal J Cohen. Catastrophic interference in connectionist networks. *Psychology of Learning and Motivation*, 24:109–165, 1989.

Timothy J O'Shea, Johnathan Corgan, and T Charles Clancy. Convolutional radio modulation recognition networks. *arXiv preprint arXiv:1602.04105*, 2016.

Timothy J O'Shea, Tamoghna Roy, and T Charles Clancy. Over-the-air deep learning based radio signal classification. *IEEE Journal of Selected Topics in Signal Processing*, 12(1):168–179, 2018.